Status	Finished
Started	Tuesday, 20 October 2026, 10:11 PM
Completed	Tuesday, 20 October 2026, 10:15 PM
Duration	4 mins 18 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the process of Merge Sort?

- a. It compares adjacent elements and swaps them if necessary
- b. It divides the list into two halves, sorts each half, and then merges them
- c. It builds a sorted array one element at a time
- d. It repeatedly finds the minimum element and moves it to the sorted part of the list

The correct answer is: It divides the list into two halves, sorts each half, and then merges them

Question **2**

Complete

Mark 1.00 out of 1.00

What is a characteristic of the merge sort algorithm?

- a. It does not require recursion
- b. It is less efficient than bubble sort
- c. It sorts data using a single pass
- d. It is based on the divide-and-conquer approach

The correct answer is: It is based on the divide-and-conquer approach

Question **3**

Complete

Mark 1.00 out of 1.00

What does the Bubble Sort algorithm primarily focus on during each pass?

- a. Bubbling up the largest element to its correct position
- b. Bubbling up the smallest element
- c. Dividing the list into halves
- d. Sorting the entire list in one pass

The correct answer is: Bubbling up the largest element to its correct position

Question **4**

Complete

Mark 1.00 out of 1.00

_____is putting an element in the appropriate place in a sorted list yields a larger sorted order list.

- a. Extraction
- b. Selection
- c. Distribution
- d. Insertion

The correct answer is:
Insertion

Question **5**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is known for its simplicity and sequential comparison of elements?

- a. Quick Sort
- b. Bubble Sort
- c. Heap Sort
- d. Merge Sort

The correct answer is: Bubble Sort

Question **6**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after the two halves of the list are sorted?

- a. They are combined to form a single sorted list
- b. They are split again into smaller sublists
- c. They are discarded
- d. They are compared element by element

The correct answer is: They are combined to form a single sorted list

Question **7**

Complete

Mark 1.00 out of 1.00

What is one advantage of sorting a list before performing a search operation?

a. It has no effect on the search operation
b. It allows for faster searching
c. It makes the search operation slower
d. It increases the number of comparisons needed

The correct answer is: It allows for faster searching

Question **8**

Complete

Mark 1.00 out of 1.00

What is one of the key advantages of using the built-in sorted() function in Python?

a. It requires external libraries
b. It sorts data out of the box efficiently
c. It is less efficient than custom sorting algorithms
d. It only works with integer arrays

The correct answer is: It sorts data out of the box efficiently

Question **9**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm involves comparing elements and swapping adjacent items that are out of order?

a. Binary Search
b. Merge Sort
c. Bubble Sort
d. Linear Search

The correct answer is: Bubble Sort

Question **10**

Complete

Mark 1.00 out of 1.00

What is the primary benefit of using sorting algorithms in programming?

- a. Provides a basis for other algorithms to work efficiently
- b. Makes data harder to manage
- c. Makes code execution slower
- d. Decreases the efficiency of algorithms

The correct answer is: Provides a basis for other algorithms to work efficiently

Question **11**

Complete

Mark 1.00 out of 1.00

What is the primary advantage of the divide-and-conquer approach in sorting algorithms?

- a. It simplifies the sorting process by using only one pass
- b. It allows for efficient parallel processing and sorting of data
- c. It only works on small datasets
- d. It avoids the need for recursion

The correct answer is: It allows for efficient parallel processing and sorting of data

Question **12**

Complete

Mark 1.00 out of 1.00

Which of the following is a key reason for the importance of sorting algorithms?

- a. Sorting decreases the efficiency of selection operations
- b. Sorting helps in finding duplicates quickly
- c. Sorting makes it harder to search for items
- d. Sorting is rarely used in programming

The correct answer is: Sorting helps in finding duplicates quickly

Question **13**

Complete

Mark 1.00 out of 1.00

_____ explain how an algorithm will perform when the input grows larger.

a. Sorting

b. Merging

c. Complexity

d. Searching

The correct answer is:
Complexity

Question **14**

Complete

Mark 1.00 out of 1.00

Which algorithm typically follows a divide-and-conquer structure?

a. Bubble Sort

b. Binary Search

c. Merge Sort

d. Linear Search

The correct answer is: Merge Sort

Question **15**

Complete

Mark 1.00 out of 1.00

Which algorithm is efficient for analyzing the frequency distribution of items in a list?

a. Bubble Sort

b. Quick Sort

c. Linear Search

d. Merge Sort

The correct answer is: Merge Sort